OpenXAI Benchmark — Full Experiment Comparison

Phase 1 & 2: Baseline Replication | Phase 3: Noise Study | Smoothing Fix

Datasets:	Adult Income | COMPAS Recidivism
Models:	Pretrained ANN + Logistic Regression
Explainers:	LIME SHAP VanillaGrad SmoothGrad IntGrad Grad x Input Random
Metrics:	PGF (Faithfulness, higher better) | RIS (Stability, lower better)
Noise Levels:	sigma = 0.0 (clean) 0.1 (mild) 0.3 (moderate) 0.5 (heavy)
Smoothing:	SmoothSHAP & SmoothLIME (K = 20 averaging copies per input)

SECTION 1

Phase 1 & 2 — Baseline Results

Clean data evaluation on Adult Income + COMPAS with ANN and LR models

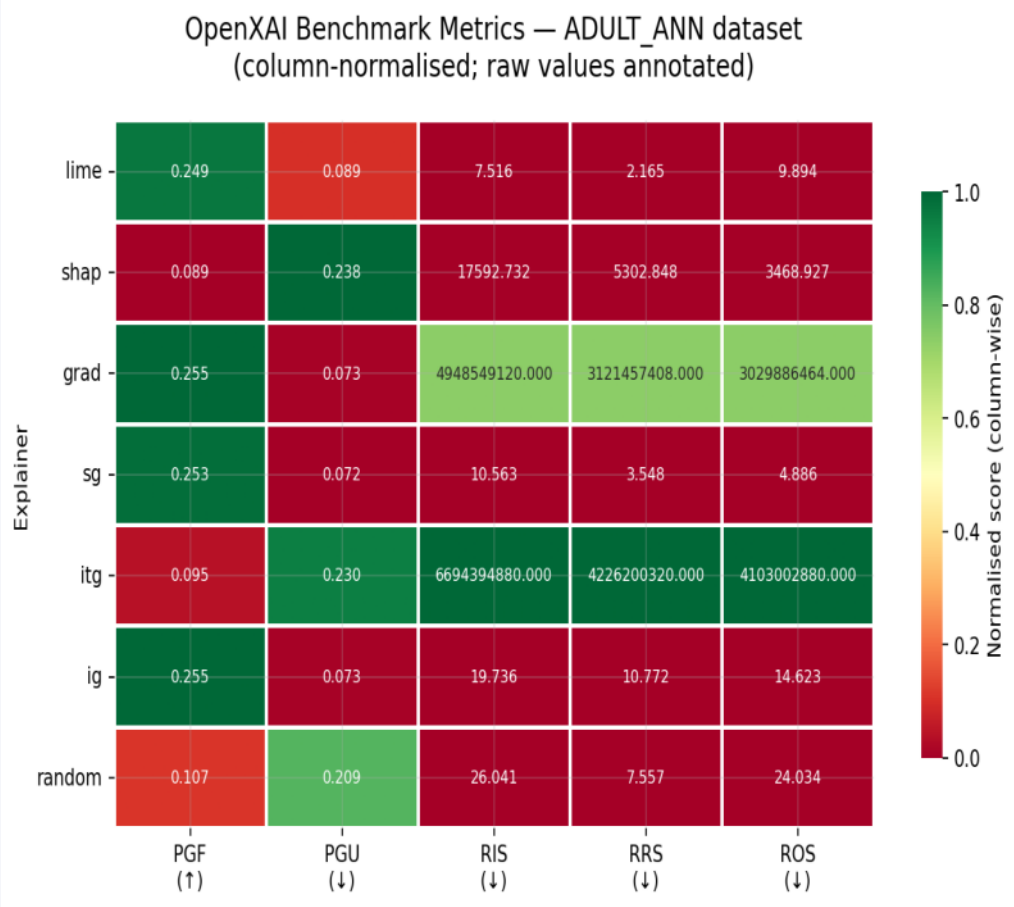
Phase 2 — Adult Income (ANN, Clean Data, sigma=0.0) — All 5 Metrics

	PGF (Higher)	PGU (Lower)	RIS (Lower)	RRS (Lower)	ROS (Lower)
LIME	0.2493	0.0892	7.5165	2.1646	9.8944
SHAP	0.0891	0.2384	17,593	5,303	3,469
VanillaGrad	0.2548	0.0733	4.95e+09	3.12e+09	3.03e+09
SmoothGrad	0.2527	0.0724	10.5627	3.5479	4.8857
Grad x Input	0.0955	0.2303	6.69e+09	4.23e+09	4.10e+09
IntGrad	0.2547	0.0734	19.7356	10.7722	14.6233
Random	0.1071	0.2087	26.0409	7.5566	24.0339

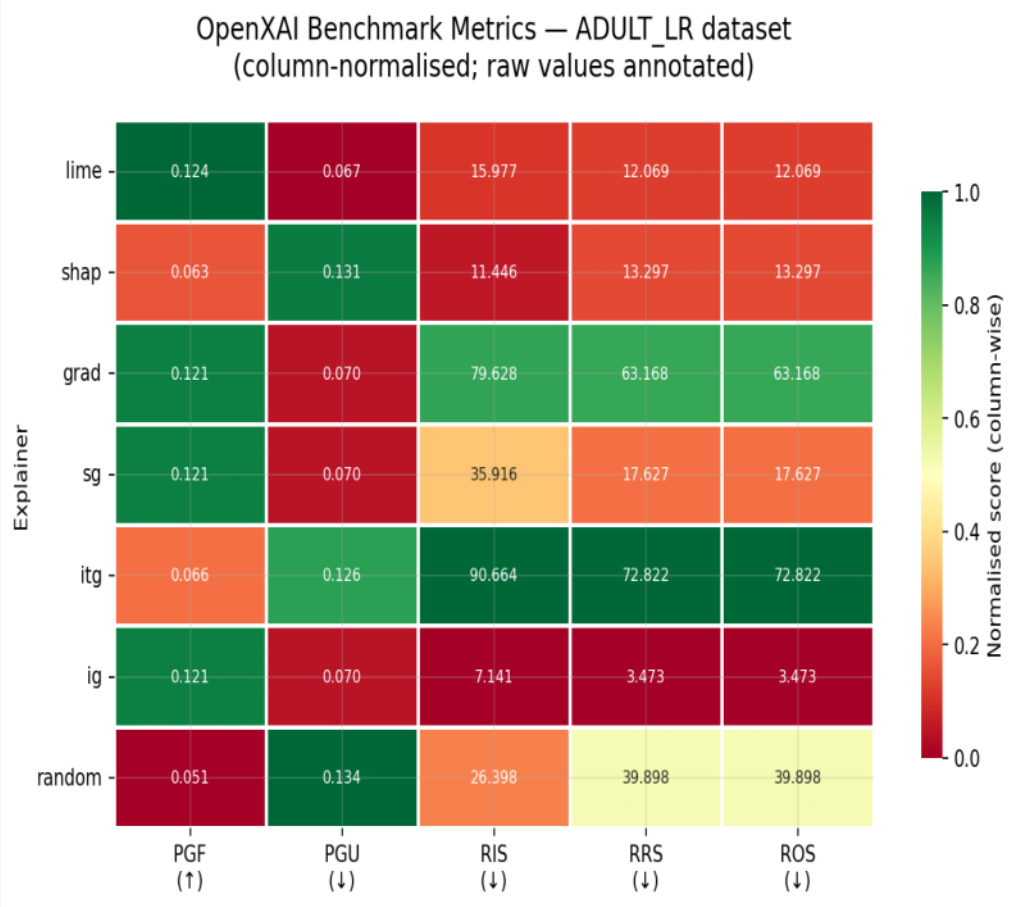
Green = good performance | Amber = moderate | Red = poor (PGF col: higher better, RIS col: lower better)

Phase 2 — Metric Heatmaps: Adult Income (ANN vs LR)

Adult Income — ANN Heatmap

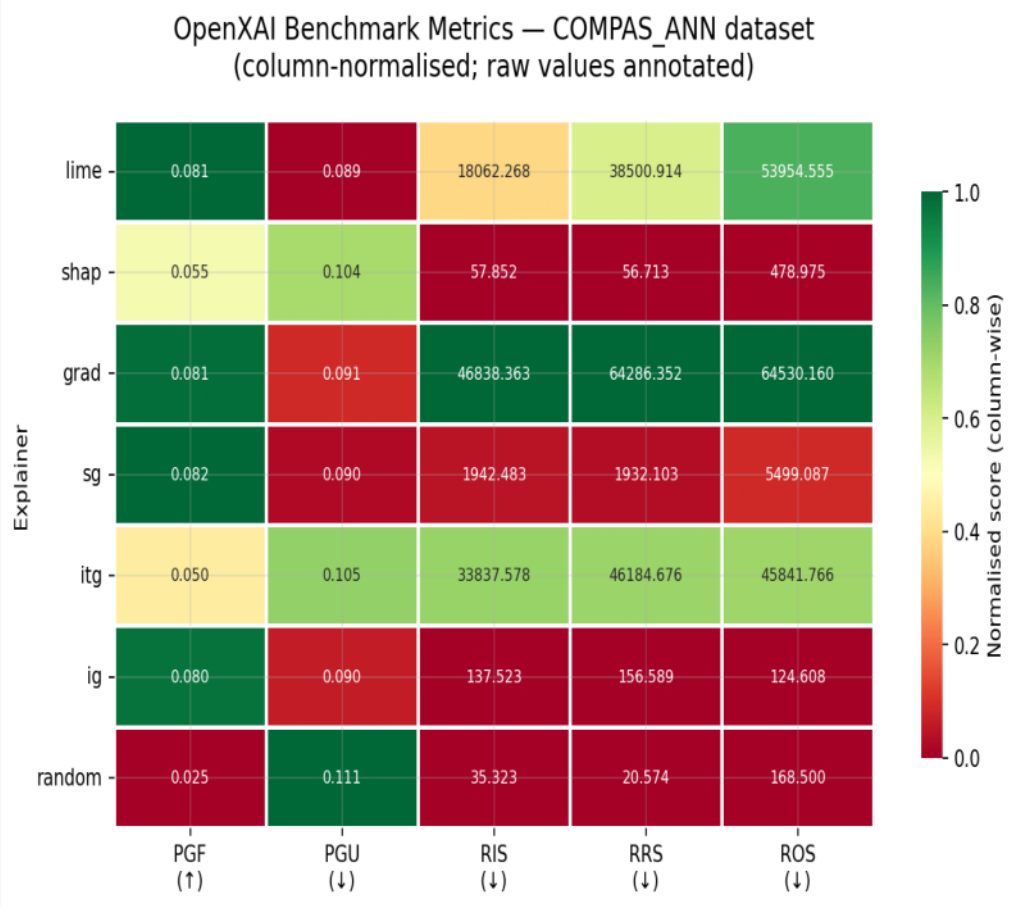


Adult Income — LR Heatmap

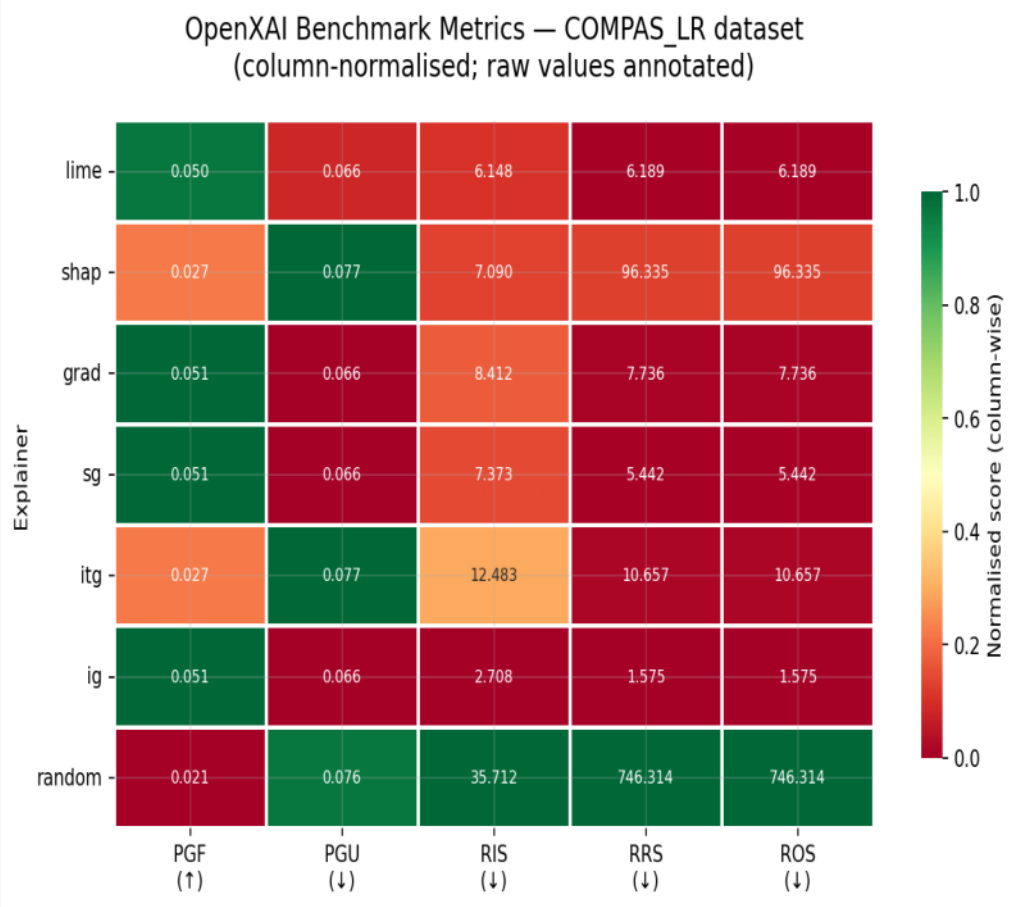


Phase 2 — Metric Heatmaps: COMPAS (ANN vs LR)

COMPAS — ANN Heatmap

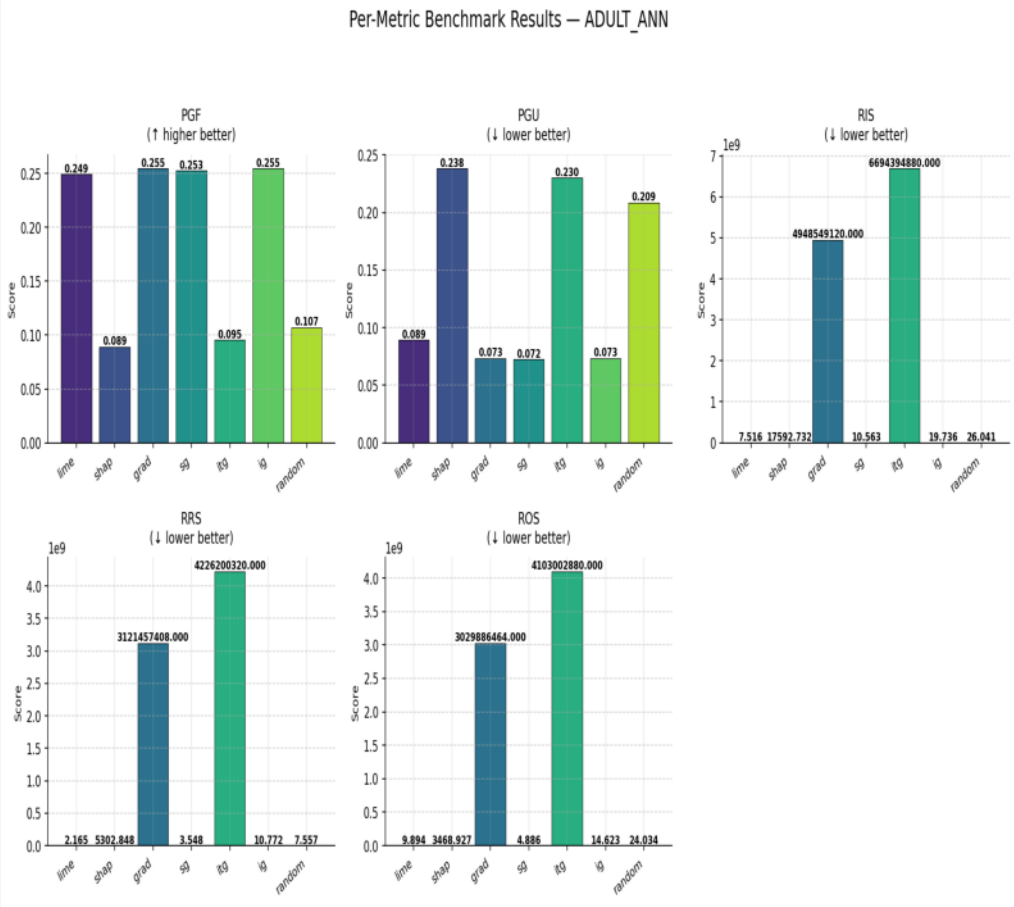


COMPAS — LR Heatmap

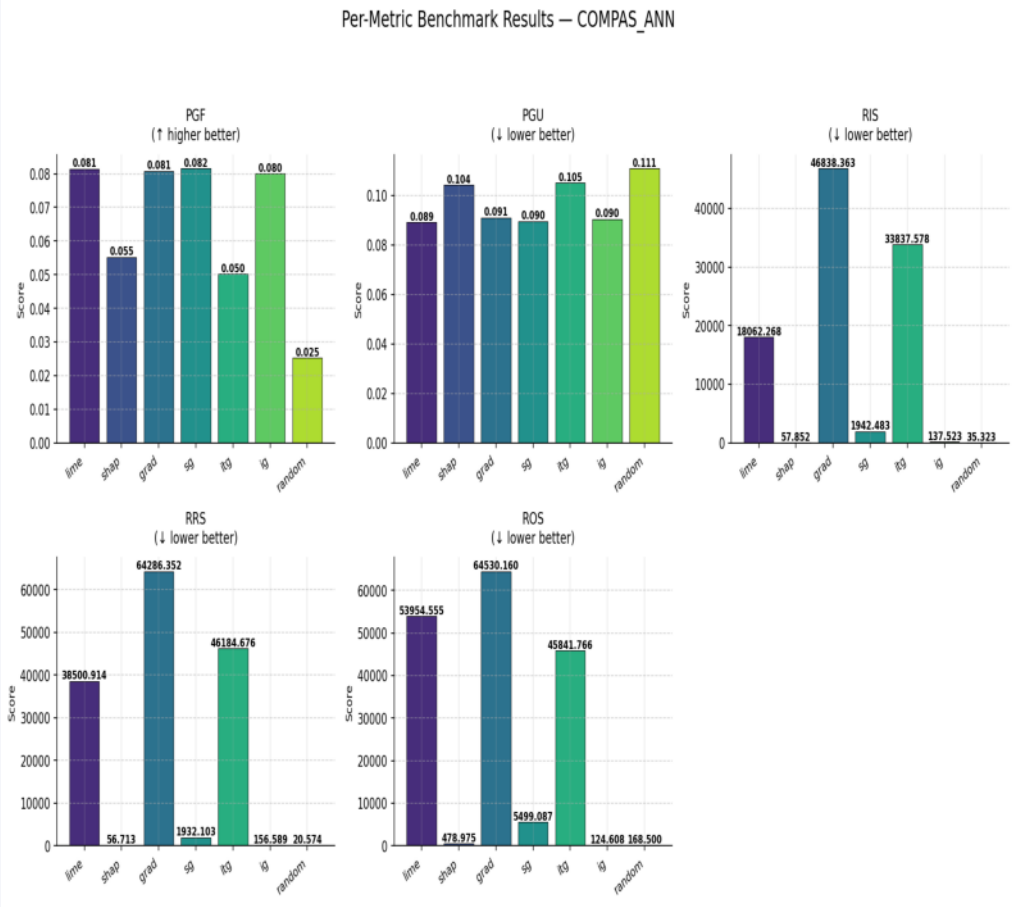


Phase 2 — Per-Metric Bar Charts (ANN Model)

Adult Income — ANN Bar Charts



COMPAS — ANN Bar Charts



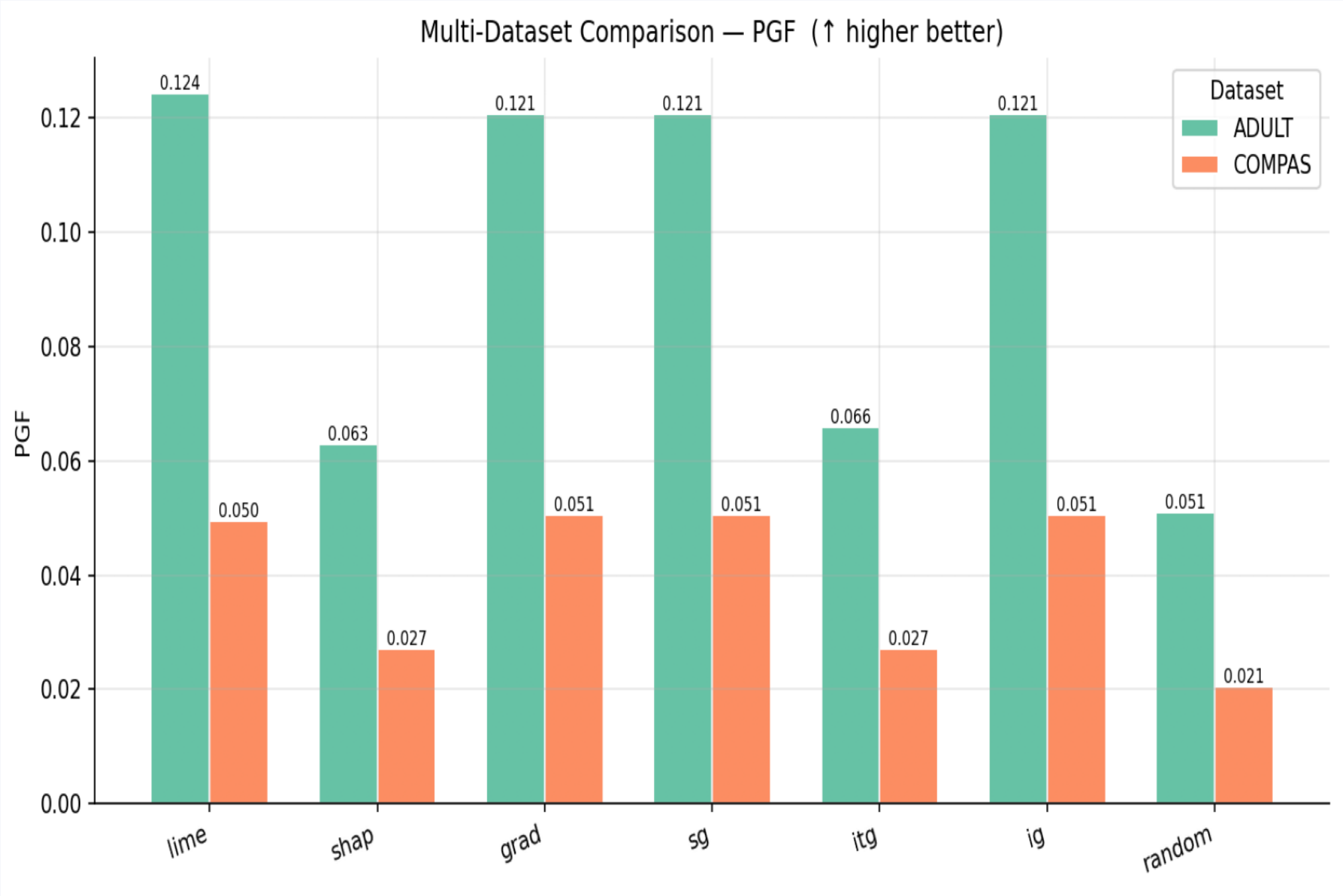


Figure: Cross-Dataset PGF Comparison — Adult Income vs COMPAS (ANN)

SECTION 2

Phase 3 Study 1 — Degradation Under Noise

All 7 explainers stressed across sigma = 0.0 to 0.5 | How bad does faithfulness and stability get?

Phase 3 — PGF (Faithfulness) Across All Noise Levels — Adult Income (ANN)

	sigma=0.0 (Clean)	sigma=0.1 (Mild)	sigma=0.3 (Moderate)	sigma=0.5 (Heavy)
LIME	0.2459	0.2761	0.2035	0.1559
SHAP	0.0932	0.1219	0.0869	0.0565
VanillaGrad	0.2488	0.2779	0.2059	0.1572
SmoothGrad	0.2474	0.2770	0.2050	0.1570
Grad x Input	0.0804	0.1364	0.0894	0.0556
IntGrad	0.2497	0.2777	0.2057	0.1566
Random	0.0869	0.1010	0.0697	0.0591

PGF = faithfulness (higher is better). Green >= 0.18 | Amber >= 0.07 | Red < 0.07

Phase 3 — PGF (Faithfulness) Across All Noise Levels — COMPAS (ANN)

	sigma=0.0 (Clean)	sigma=0.1 (Mild)	sigma=0.3 (Moderate)	sigma=0.5 (Heavy)
LIME	0.0866	0.0890	0.0858	0.0743
SHAP	0.0676	0.0652	0.0491	0.0322
VanillaGrad	0.0856	0.0884	0.0864	0.0743
SmoothGrad	0.0881	0.0902	0.0885	0.0761
Grad x Input	0.0647	0.0643	0.0468	0.0360
IntGrad	0.0861	0.0887	0.0874	0.0759
Random	0.0318	0.0292	0.0323	0.0216

PGF = faithfulness (higher is better). Green >= 0.18 | Amber >= 0.07 | Red < 0.07

Phase 3 Study 1 — Degradation: Adult Income ANN
Clean (sigma=0.0) vs Heavy Noise (sigma=0.5)

	PGF Clean	PGF Sigma=0.5	PGF Drop %	RIS Clean	RIS Sigma=0.5	RIS Change
LIME	0.2459	0.1559	-36.6%	12.5863	508,515	x40402.2
SHAP	0.0932	0.0565	-39.4%	71,467	258,524	x3.6
VanillaGrad	0.2488	0.1572	-36.8%	1.02e+12	1.29e+14	x126.3
SmoothGrad	0.2474	0.1570	-36.5%	109.8349	88,262,984	x803596.6
Grad x Input	0.0804	0.0556	-30.9%	1.37e+12	1.35e+14	x98.8
IntGrad	0.2497	0.1566	-37.3%	19.4804	12.7628	x0.7
Random	0.0869	0.0591	-32.0%	25.7540	25.6573	x1.0

PGF: faithfulness (higher = better) | RIS: stability (lower = better) | Drop% = how much faithfulness fell | xN = how many times RIS worsened

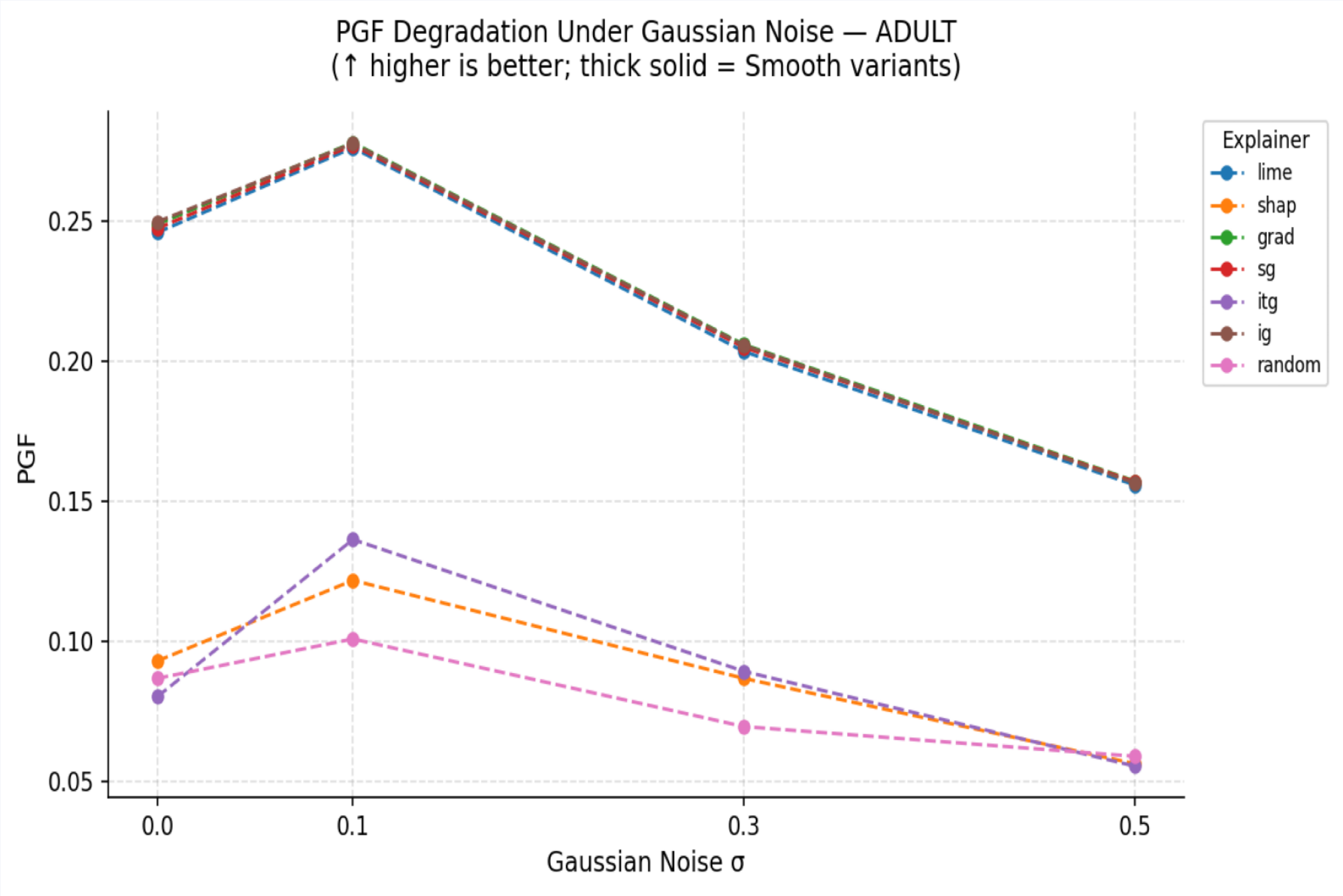


Figure: Adult Income — PGF (Faithfulness) Degradation Curves across all sigma levels

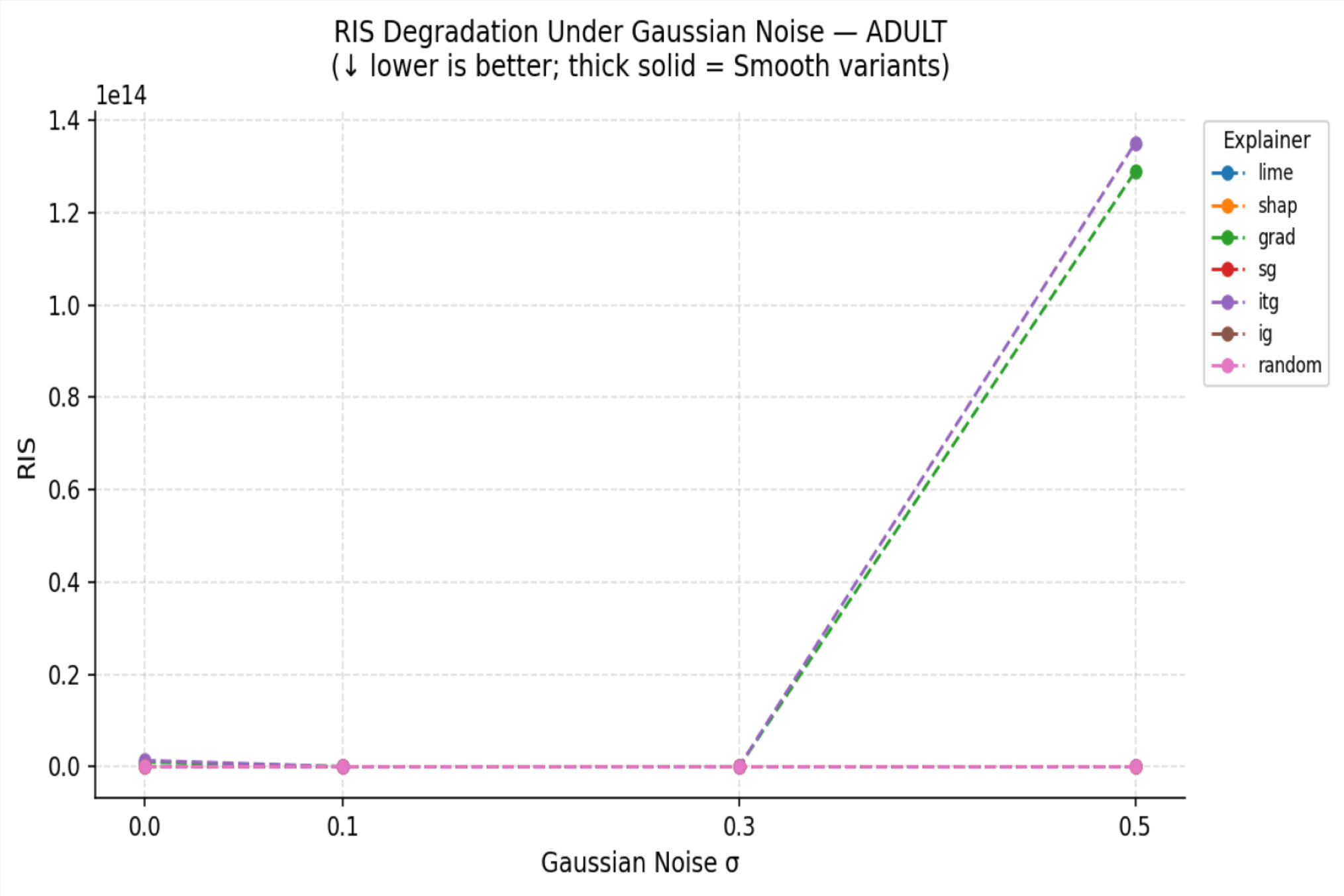


Figure: Adult Income — RIS (Stability) Degradation Curves across all sigma levels

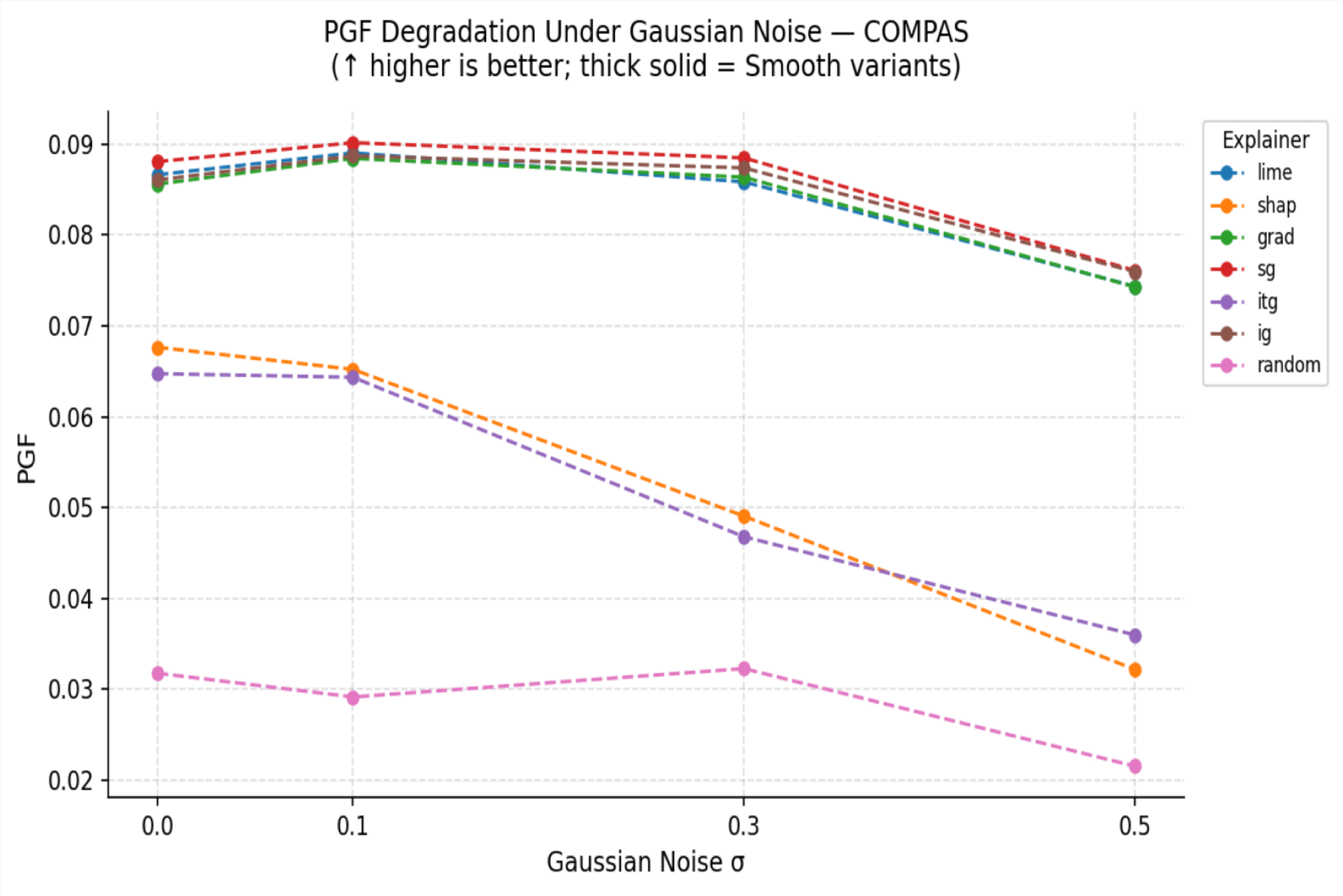


Figure: COMPAS — PGF (Faithfulness) Degradation Curves across all sigma levels

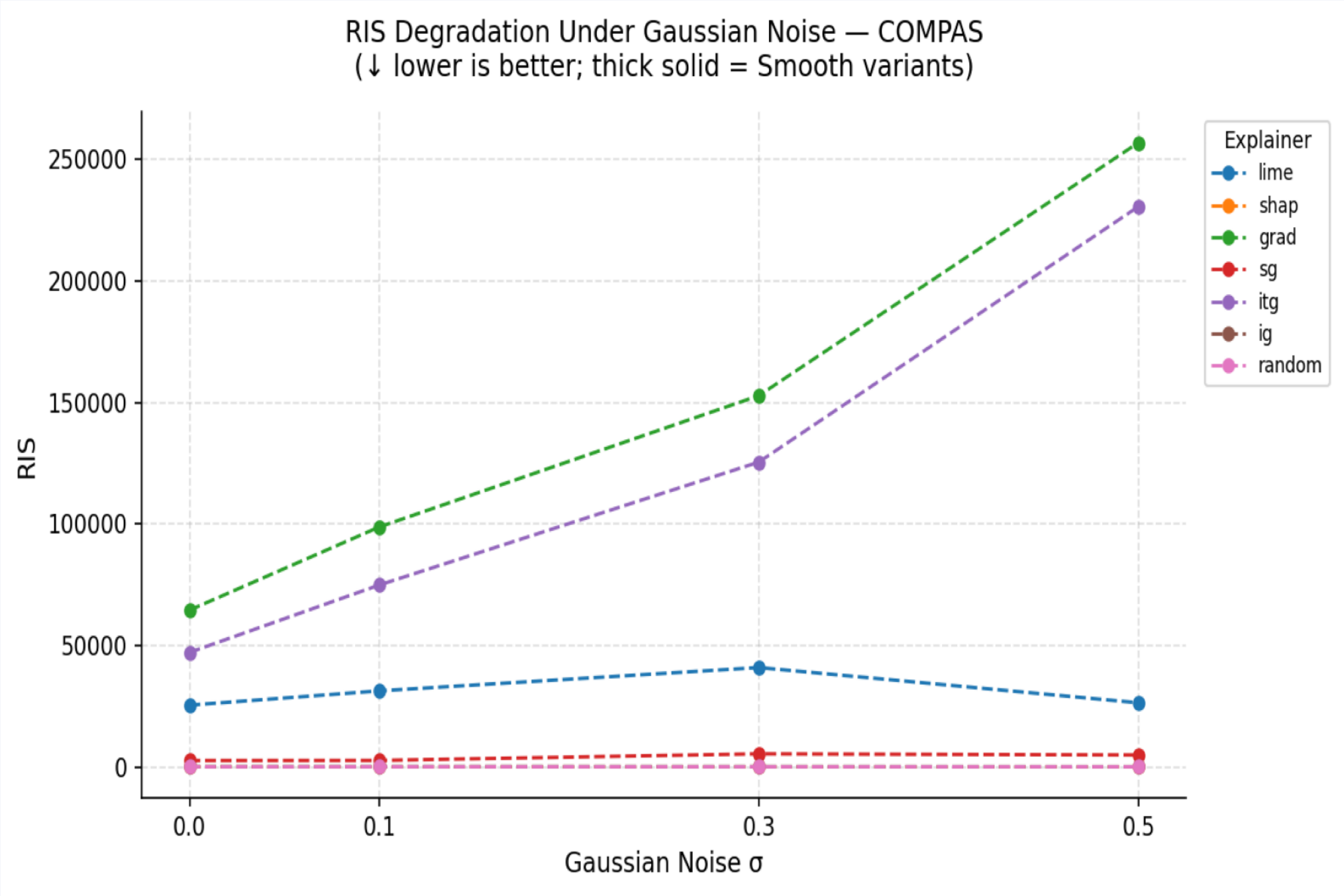
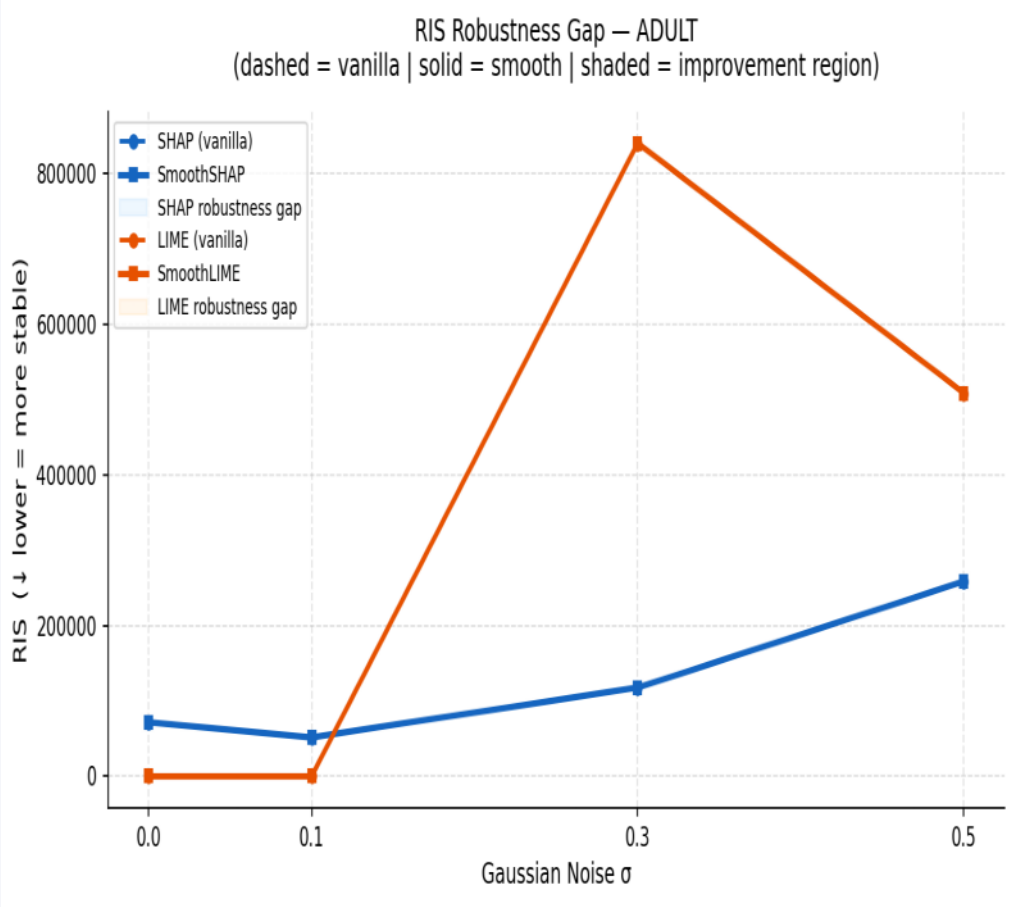


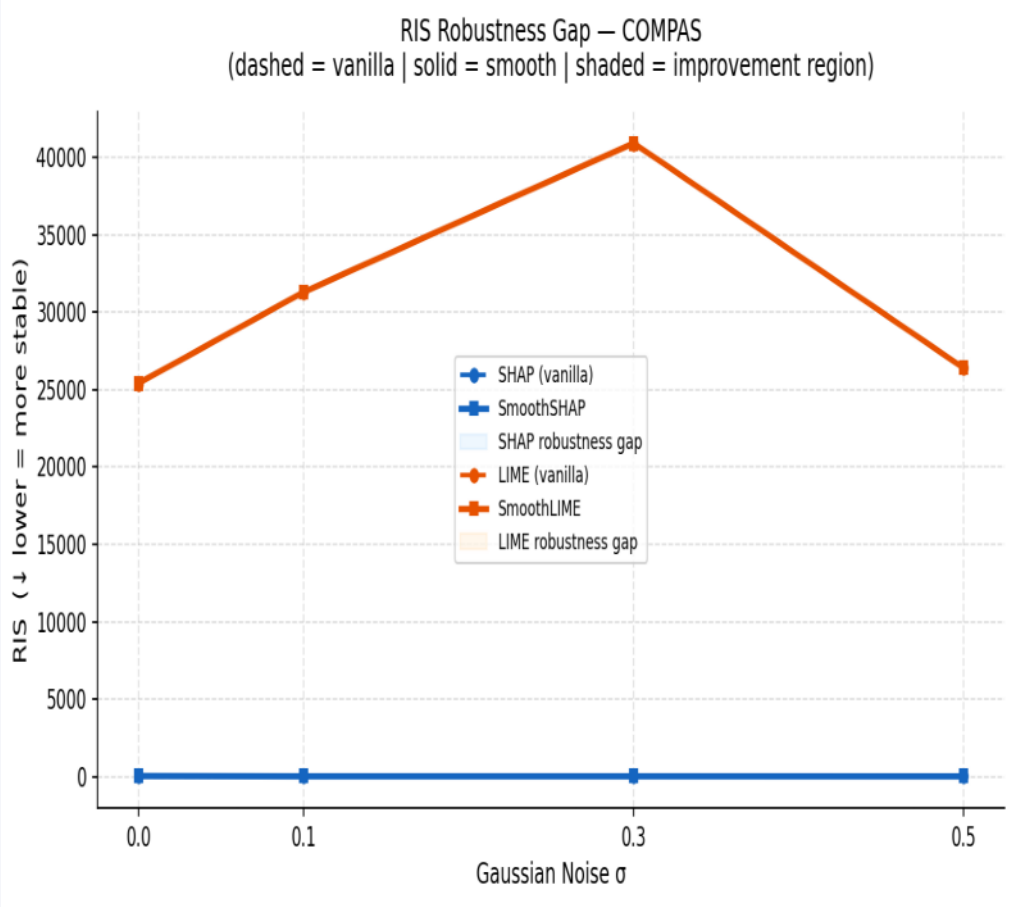
Figure: COMPAS — RIS (Stability) Degradation Curves across all sigma levels

Phase 3 — Robustness Gap Between Best and Worst Explainer

Adult Income — Robustness Gap



COMPAS — Robustness Gap



SECTION 3

Phase 3 Study 2 — The Smoothing Fix

SmoothSHAP and SmoothLIME vs vanilla versions across all noise levels | Can averaging K=20 copies recover explanation quality?

Phase 3 Study 2 — Smoothing Fix: Vanilla vs Smooth Explainers — Adult Income

SHAP vs SmoothSHAP

Sigma	Vanilla PGF	Smooth PGF	Gain %	Verdict
sigma=0.0	0.0906	0.0840	-7.3%	No change
sigma=0.1	0.1219	0.2195	+80.1%	Big win!
sigma=0.3	0.0869	0.2026	+133.1%	Big win!
sigma=0.5	0.0565	0.1537	+172.1%	Big win!

LIME vs SmoothLIME

Sigma	Vanilla PGF	Smooth PGF	Gain %	Verdict
sigma=0.0	0.2462	0.2479	+0.7%	Marginal
sigma=0.1	0.2761	0.2767	+0.2%	Marginal
sigma=0.3	0.2035	0.2038	+0.2%	Marginal
sigma=0.5	0.1559	0.1562	+0.2%	Marginal

Phase 3 Study 2 — Smoothing Fix: Vanilla vs Smooth Explainers — COMPAS

SHAP vs SmoothSHAP

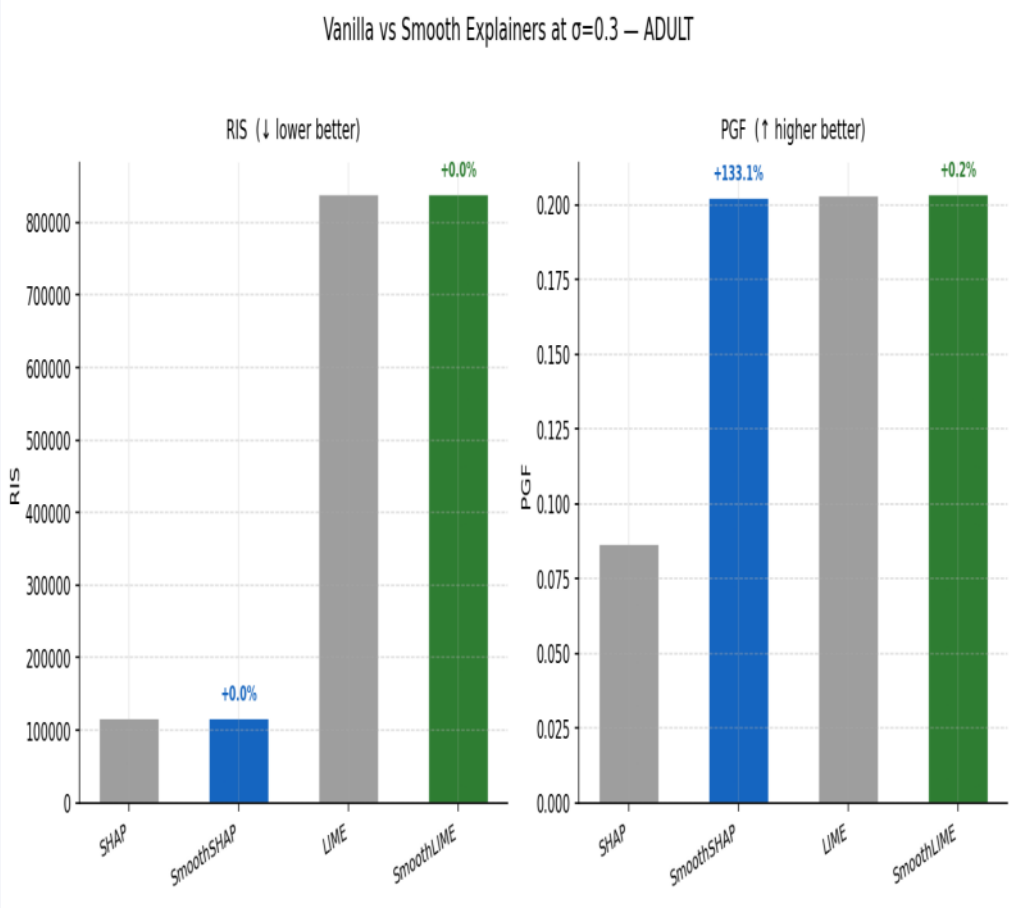
Sigma	Vanilla PGF	Smooth PGF	Gain %	Verdict
sigma=0.0	0.0674	0.0666	-1.2%	No change
sigma=0.1	0.0652	0.0710	+8.9%	Improved
sigma=0.3	0.0491	0.0701	+42.9%	Big win!
sigma=0.5	0.0322	0.0593	+84.0%	Big win!

LIME vs SmoothLIME

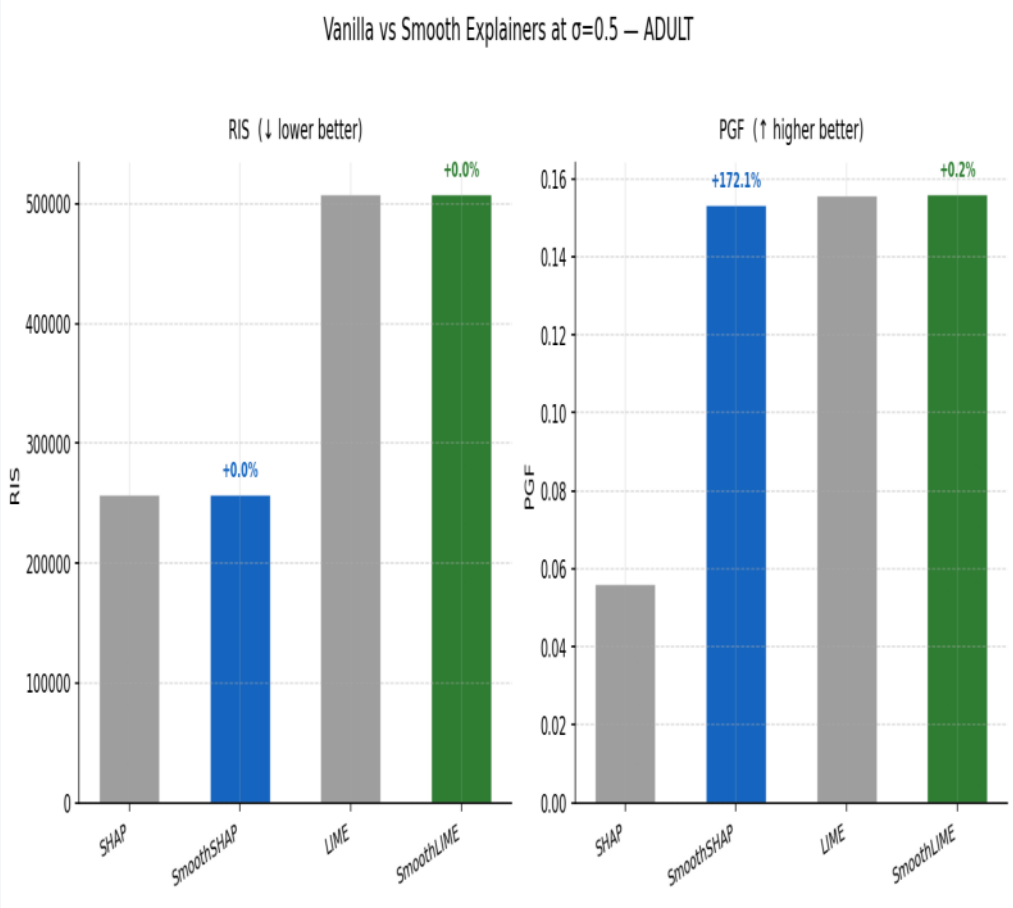
Sigma	Vanilla PGF	Smooth PGF	Gain %	Verdict
sigma=0.0	0.0880	0.0880	-0.1%	No change
sigma=0.1	0.0890	0.0901	+1.2%	Marginal
sigma=0.3	0.0858	0.0881	+2.6%	Marginal
sigma=0.5	0.0743	0.0752	+1.2%	Marginal

Phase 3 — Before vs After Smoothing: Adult Income

Adult — Before vs After sigma=0.3

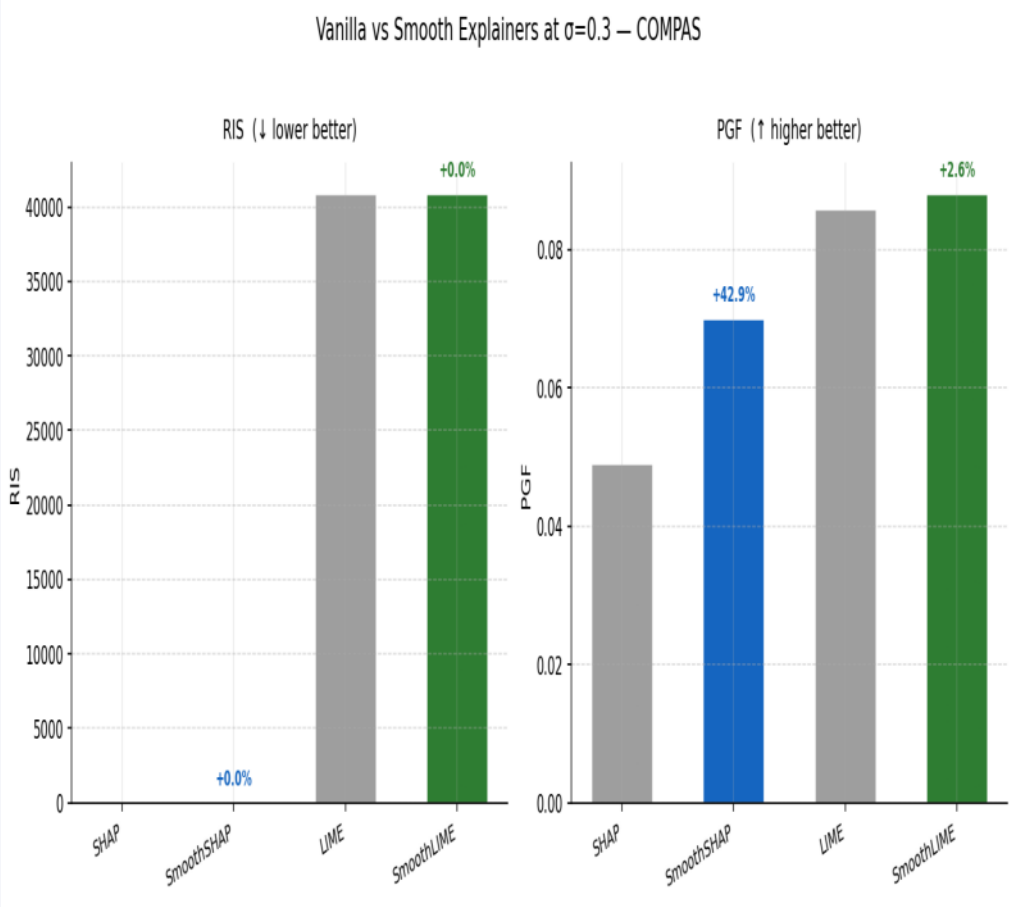


Adult — Before vs After sigma=0.5

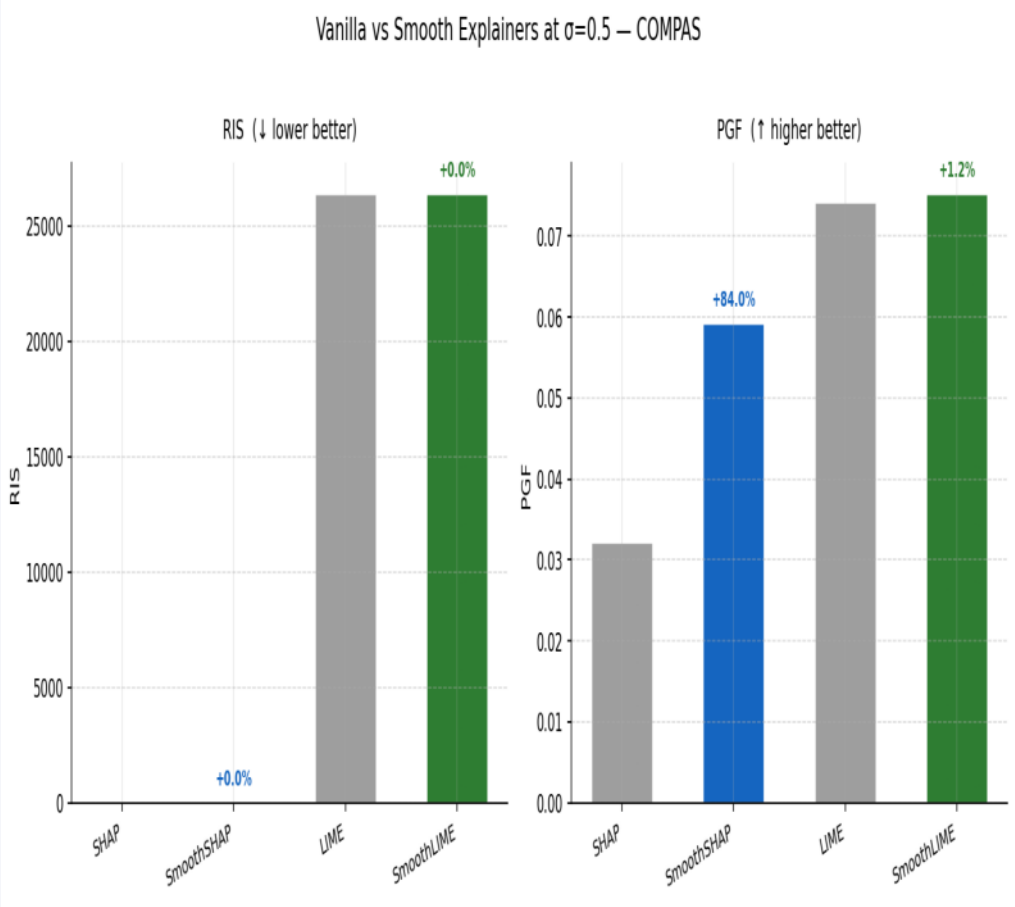


Phase 3 — Before vs After Smoothing: COMPAS

COMPAS — Before vs After sigma=0.3



COMPAS — Before vs After sigma=0.5



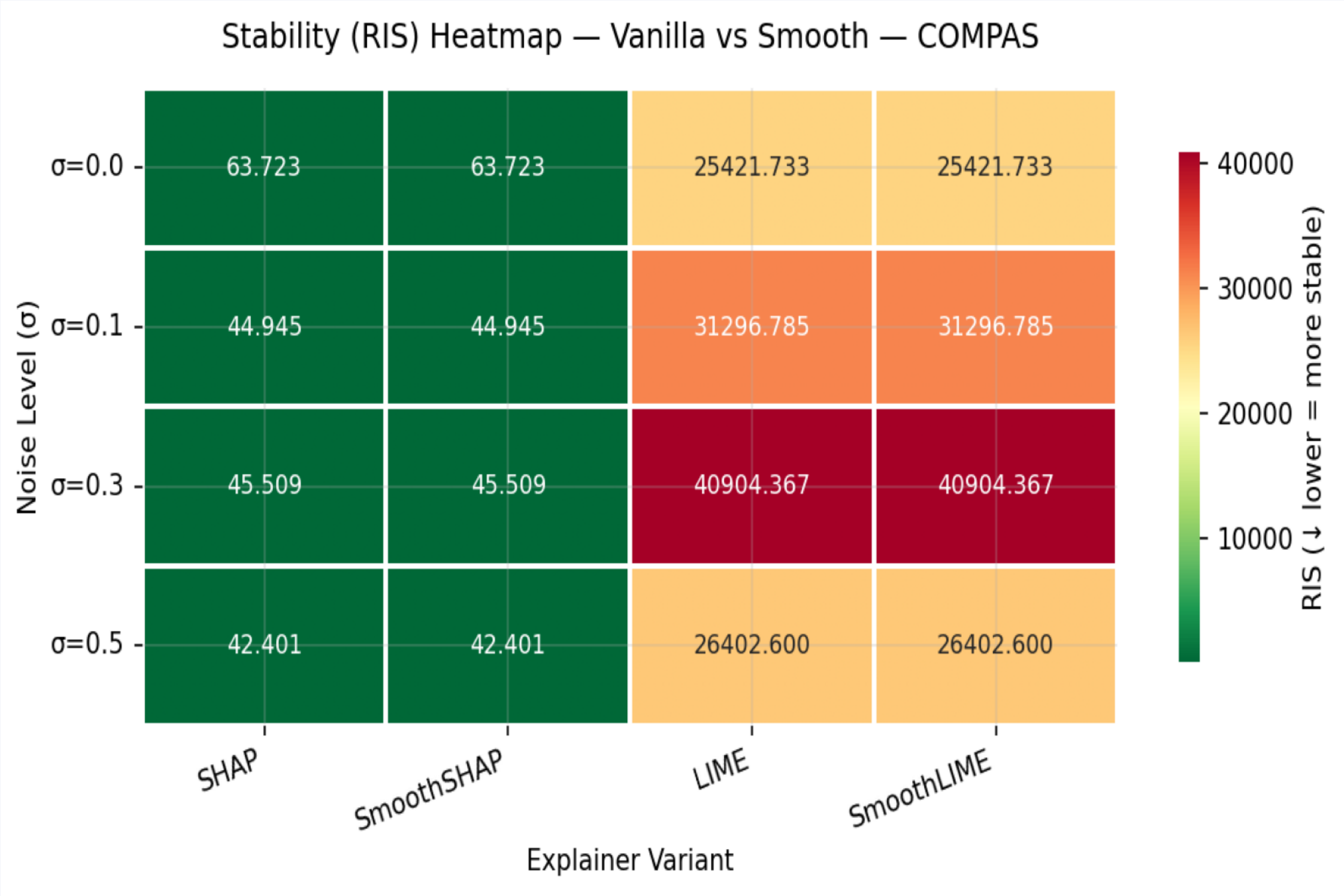


Figure: Smoothing Benefit Heatmap — delta PGF (SmoothSHAP / SmoothLIME vs Vanilla)

SECTION 4

Final Verdict

Which explainer is best — and under what conditions?

Final Verdict — Summary of All Conditions

Explainer	Clean Faithfulness	Clean Stability	At Sigma=0.5 Faithfulness	Smoothing Fix Works?	Recommendation
IntGrad (ig)	[+] High	[+] Best	[+] Holds best	N/A	Best overall on Adult
SmoothGrad (sg)	[+] High	[~] Moderate	[!] Collapses	N/A	Good on clean data only
LIME	[+] High	[+] Good	[~] -37% drop	[~] Marginal	Reliable, but slow
SHAP	[~] Medium	[!] Unstable	[!] -40% drop	[+] +172%	Use SmoothSHAP in noisy e
VanillaGrad	[+] High	[!] Explodes	[!] RIS ->10^14	N/A	Avoid for stability
Grad x Input	[~] Medium	[!] Worst	[!] Worst	N/A	Avoid
Random	[!] Floor	[+] Stable	[+] Consistent	N/A	Baseline reference only

[+] = Good | [~] = Moderate | [!] = Poor/Dangerous